MyoDishAnalysis: pulse assignment and settings guide

Markers of the force plot, assignment of stimulus pulses to contractions, and the meaning of every advanced setting

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How to use this guide

- All parameters of the method that are based on assumptions or experience are **advanced settings**: GUI **Advanced ...** (one tab per part of the analysis), option names of *mda_options* (MATLAB) and *options()* (Python). The defaults are the former constants, except diastolicLevel, steadyStateBeats and prpReference (changed in 1.0.0-beta.4; the former behaviour is an option value: 'minimum', 0, 'steady') and gateMaxCertain (1.0.0-beta.5; former behaviour: 0.15 s); change them only if you understand the step concerned.
- Hover over a name in the window for a short description. Settings marked * are applied when the data are read; the window is read again.
- Every result file contains all settings (sheet / file *info*, rows option_<name>). **Save settings ...** writes them as a settings file (.csv, key / value); **Load settings ...** reads a settings file or the results of an earlier analysis. Use a settings file with *MyoDishAnalysis(..., 'settings', file)*, *mda-analyze --settings file*, the watcher, or *mda_options('settings', file)*; options given after it override it.
- Section 1 explains the markers of the force plot; sections 2–3 the assignment of stimulus pulses; sections 4–11 the settings of each tab with a figure and a table of all its parameters.

1 Markers of the force plot

The legend is shown in the force plot of both GUIs (menu *View* → *Legend of the force plot*, or right click in the force plot; it can be dragged). Entries that do not occur in the loaded window are left out.

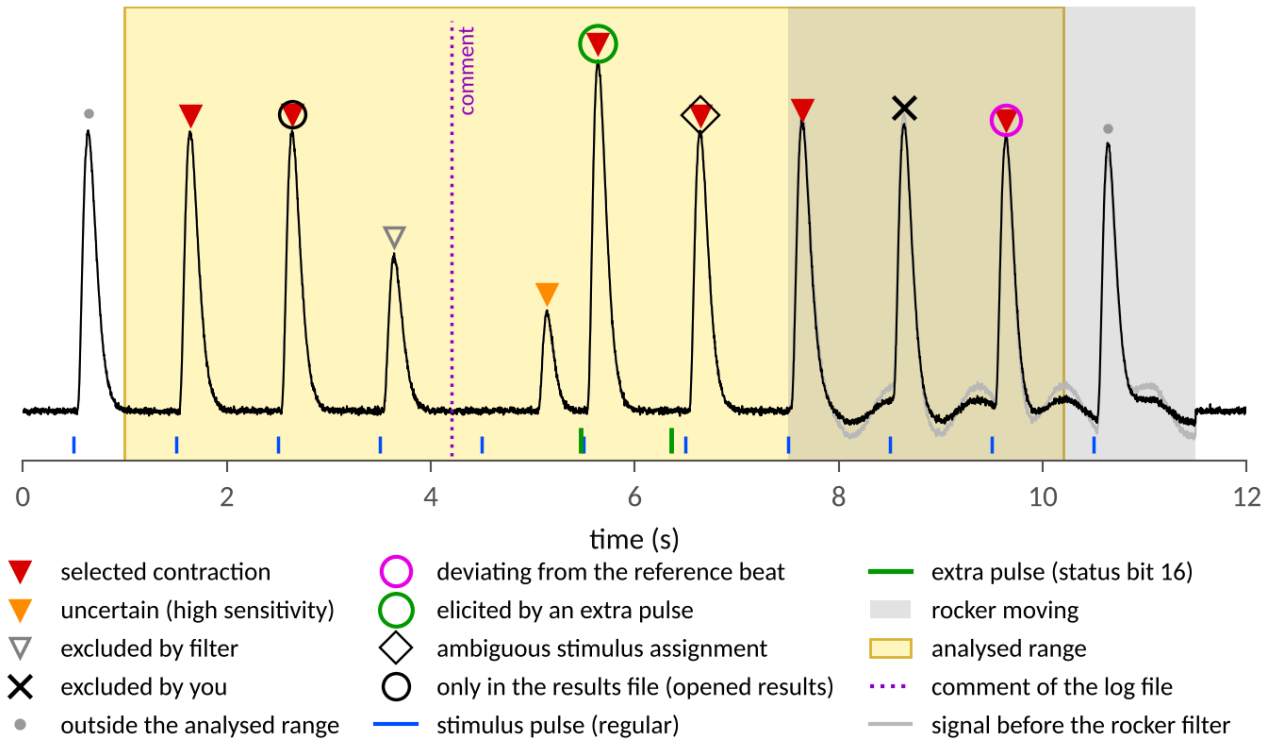


Figure 1. All markers of the force plot (schematic recording). Triangles mark the contraction peaks; rings and diamonds are added to the triangle of a contraction; ticks at the bottom mark the stimulus pulses (height of the extra pulse ticks: larger than the regular ticks).

2 Stimulus pulses and contractions

MyoDish delivers **regular pulses** (pacing) and, if defined in a schedule file, **extra pulses** (commands with the suffix #1 … #9). Every delivered pulse is written to the status channel with its time, channel and current; extra pulses carry a flag (bit 16). Pulse durations come from the log file. For the analysis every pulse is a stimulation event; regular and extra pulses differ only in time and properties.

Each contraction is linked to the pulse that elicited it, or it is an **extra beat** if no pulse can have elicited it. Each pulse either **elicits** a contraction, is **missed** (in diastole or during a contraction elicited by another pulse), or **modulates** a contraction as a **pre-pulse** before it or a **post-pulse** during it. The link is made at the **onset** of the contraction: the delay from the stimulus to the onset is short and stable across species and pacing rates (median 6–75 ms per channel in rat, rabbit, pig and human slices), the delay to the peak is not (about 50–280 ms).

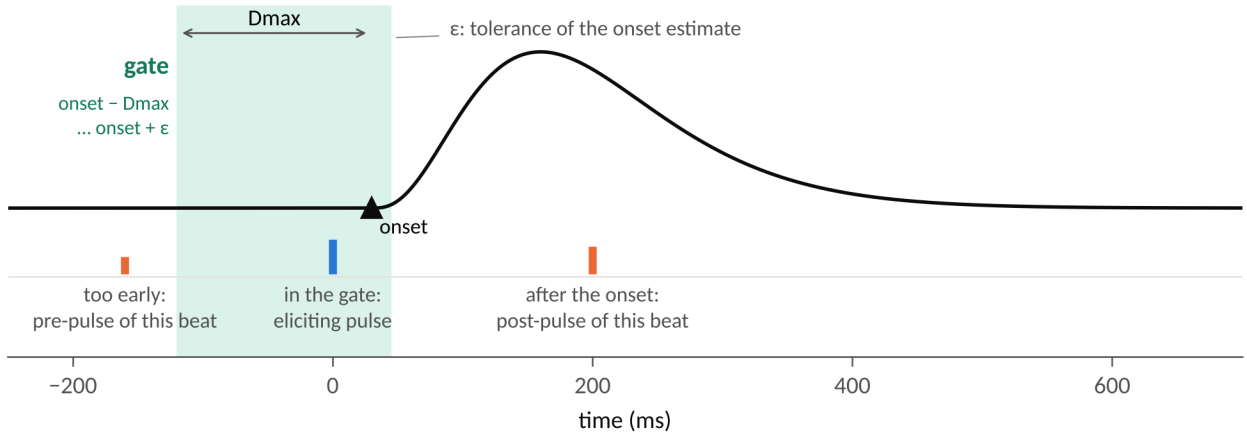


Figure 2. The gate of a contraction onset (onset − gateMax … onset + gateTolerance) and the three possible positions of a pulse relative to it.

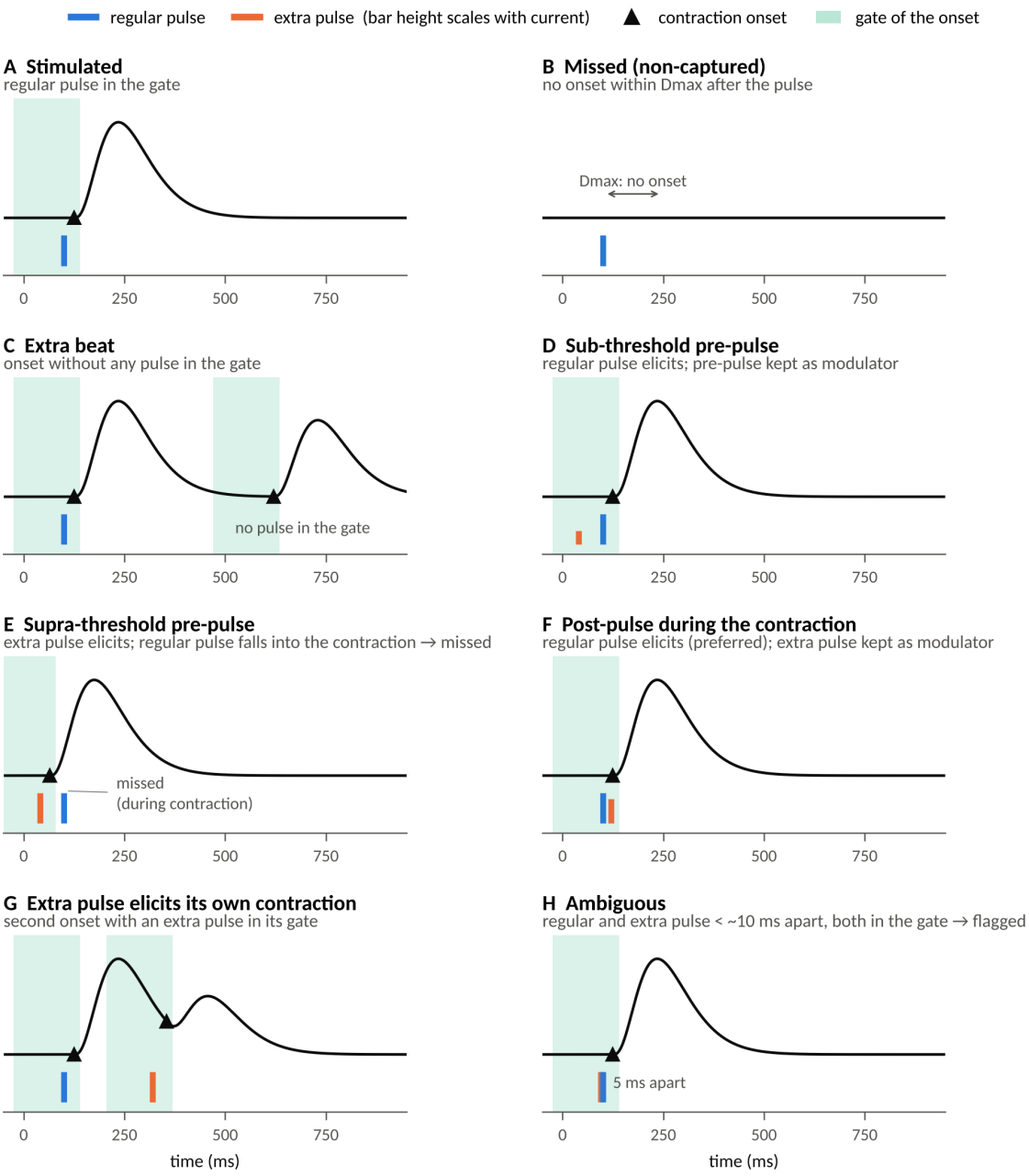


Figure 3. The cases of the assignment (schematic). Blue: regular pulse; orange: extra pulse, bar height = current (small bar: sub-threshold). Triangle: contraction onset; green: gate of the onset.

3 The assignment rules

- **Eliciting pulse.** A pulse in the gate. Several pulses: pulses before the onset are preferred over pulses after it; regular pulses over extra pulses; among pulses of one kind the earliest within gateCore before the onset, otherwise the latest one.
- **Longer gate of certain contractions** (since 1.0.0-beta.5). A certain contraction (auto threshold, paced channel) without a pulse within gateMax may be elicited by a regular pulse up to min(gateMaxCertain, gateMaxCertainRel x the interval of the regular pulses before the onset) before its onset (0.5 s, 0.5; at least gateMax), e.g. a slice with a long electromechanical delay. This second pass follows the assignment of all contractions with gateMax, so it never takes a pulse from a contraction within gateMax of it. Uncertain contractions (e.g. rocker artifacts) keep gateMax (Figure 11a).
- **Rise phases.** No pulse in the gate of the main onset: the onsets of the other rise phases are tried, the earliest first; then a pulse up to steepRiseMargin before the steepest rise (ambiguous).
- **One contraction per pulse.** Contractions are assigned in the order of their size.
- **Ambiguous** (column stimAmbiguous, black diamond): a pulse of the other kind within ambiguityWindow, a different choice with the onset gateTolerance earlier or later, or the fallback of a later rise phase / the steepest rise.
- **Pre- and post-pulses.** A pulse without own contraction is a post-pulse of the contraction it falls into (eliciting pulse … contractionEnd), otherwise a pre-pulse of the next contraction if it lies within prePulseWindow before its eliciting pulse. Text form t<ms>|<mA>|<charge>|<pause>|<discharge> (µs), several joined by &; the pulse table lists every pulse with outcome and role.
- **Peak assignment** (stimAssignment = peak): the method of versions ≤ 1.0.0-beta.3 (last pulse minStimToPeak … maxStimToPeak before the peak; extra pulses count as regular pulses).

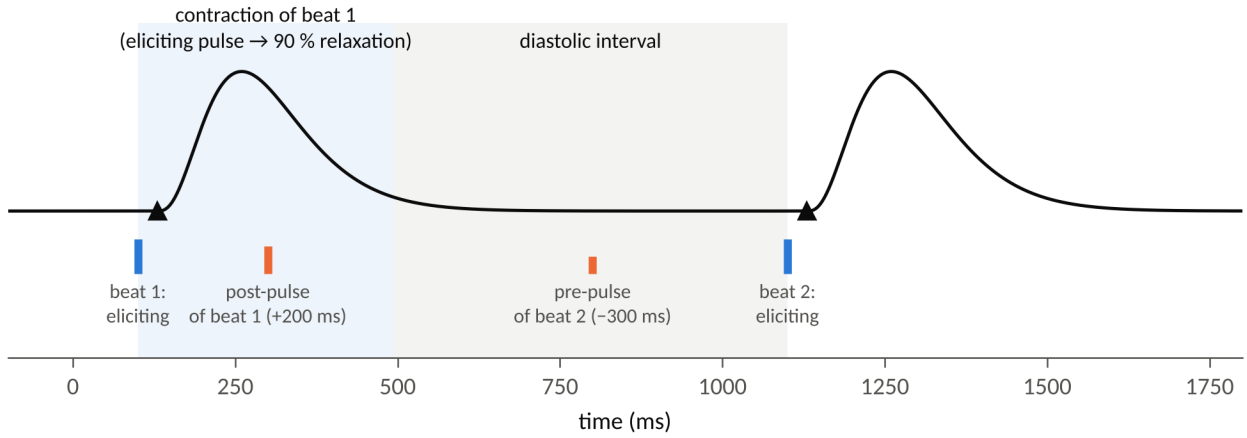


Figure 4. Pulses that do not elicit a contraction: post-pulses during a contraction (blue area), pre-pulses of the next contraction in the diastolic interval (gray area).

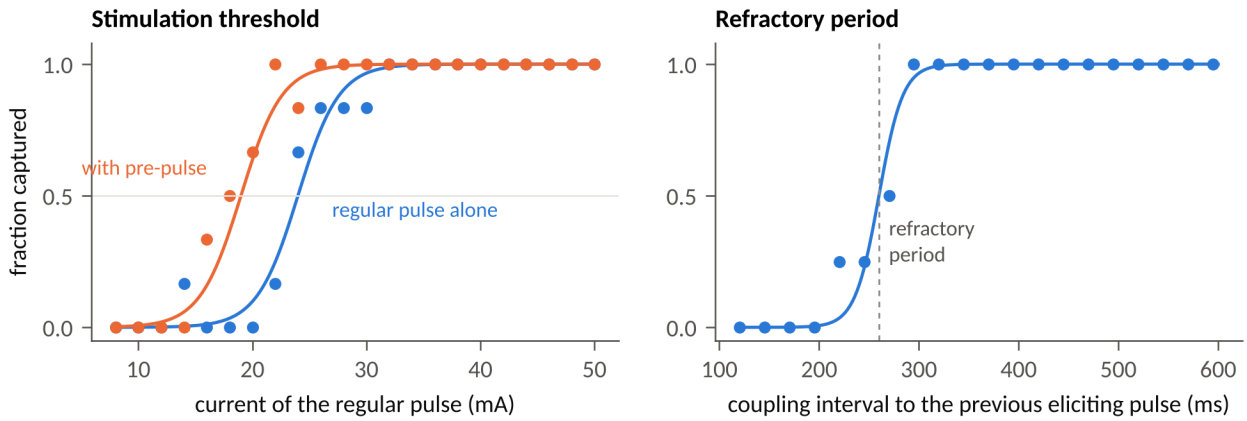


Figure 5. Stimulation threshold and refractory period as special cases of the per-pulse outcome (schematic data).

4 Tab Assignment: stimulus assignment and stimulus source

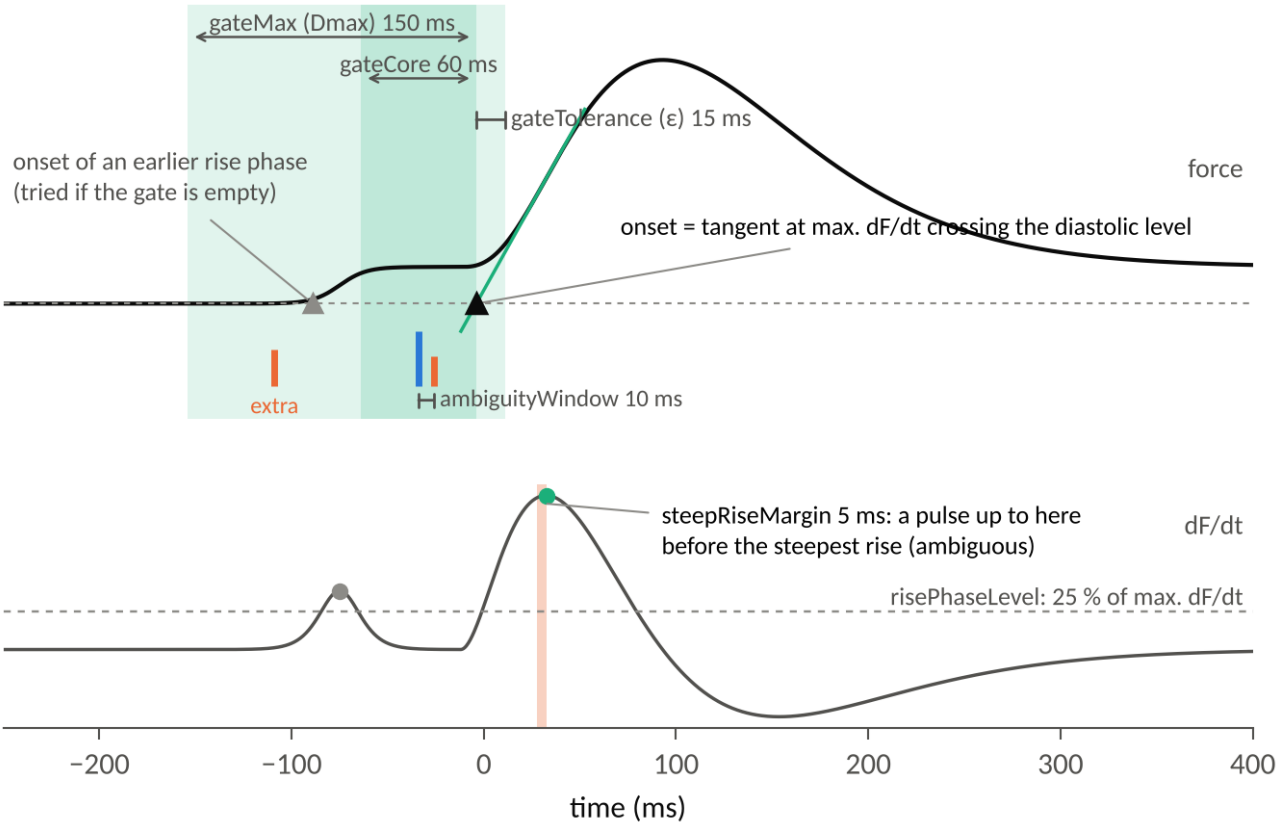


Figure 6. Onset gate and its parameters. Main onset: tangent at the maximum dF/dt (green line) crossing the diastolic level. The regular pulse lies within gateCore and is chosen; the extra pulse 8 ms after it is within ambiguityWindow (flagged); the earlier extra pulse is a pre-pulse. Lower trace: dF/dt; rise phases are local maxima ≥ risePhaseLevel x the maximum.

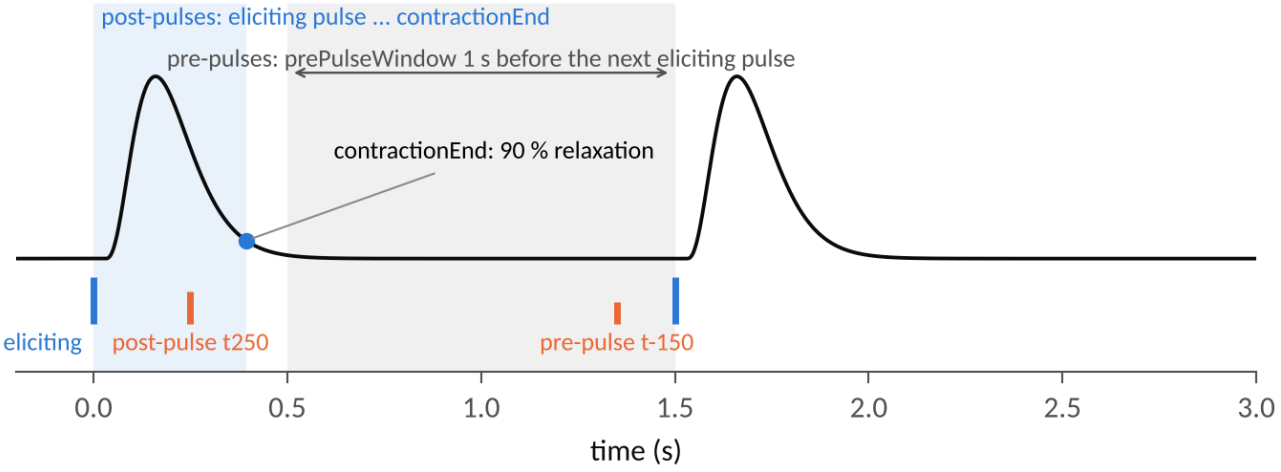


Figure 7. contractionEnd (end of a contraction for post-pulses, missed pulses during a contraction and the phase in the pulse table) and prePulseWindow.

Settings of the tab (* = the data are read again):

setting (GUI label)	default	meaning
stimAssignment Assignment	onset gate	onset gate: a pulse can only have elicited a contraction if it lies from onset - gateMax to onset + gateTolerance (onset = tangent at the maximum dF/dt crossing the diastolic level). peak: last pulse minStimToPeak … maxStimToPeak before the peak (versions <= 1.0.0-beta.3; extra pulses count as regular pulses)
gateMax longest pulse -> onset delay	150 ms	longest delay from the eliciting pulse to the contraction onset (Dmax) of uncertain contractions (and without certainty: manual threshold, unpaced). Near threshold the onset can be delayed by ~80 ms
gateMaxCertain … of certain contractions	500 ms	longest pulse -> onset delay of certain contractions (auto threshold, paced: locked to the stimuli or large); slices with a long but constant delay (e.g. 0.3 s). At most gateMaxCertainRel x the stimulus interval, at least gateMax
gateMaxCertainRel … at most	0.5 x stimulus interval	the gate of certain contractions is at most this fraction of the interval of the regular pulses before the onset (a longer delay cannot be told from an early contraction before the next pulse)
gateCore typical delay range	60 ms	several candidate pulses: pulses before the onset before pulses after it, regular before extra pulses; of one kind the earliest up to gateCore before the onset, otherwise the latest one
gateTolerance pulses after the onset	15 ms	tolerance of the onset estimate (epsilon): pulses up to this time after the onset are still candidates; ambiguous if a pulse of the other kind would be chosen with the onset this time earlier or later
ambiguityWindow	10 ms	a pulse of the other kind (regular / extra) up to this time before or after the chosen pulse marks the assignment as ambiguous (column stimAmbiguous)
risePhaseLevel rise phases	25 % of max dF/dt	no pulse in the gate of the onset: the onsets of the other rise phases of the upstroke are tried (local maxima of dF/dt of at least this fraction of the maximum dF/dt; rocker movement, fused spontaneous events)
steepRiseMargin	5 ms	still no pulse: a pulse after the gate up to this time before the maximum dF/dt is taken (onset estimated too early, e.g. rocker artifacts); marked ambiguous
contractionEnd end of a contraction	90 % relaxation	end of a contraction for post-pulses, for regular pulses missed within a contraction and for the phase of the pulse table: this relaxation (% of peak - minimum after the peak)
prePulseWindow pre-pulses kept	1 s	pulses without own contraction are listed as pre-pulses of the next contraction if they lie at most this time before its eliciting pulse (column prePulses); during a contraction (eliciting pulse … contractionEnd) as post-pulses
offsetSnap programmed offsets	5 ms	time of pre- / post-pulses: the programmed offset of the log file ('Sequence' entries) is used if the measured offset (resolution 2.5 ms) lies within this time of it
minStimToPeak peak assignment	25 ms	peak assignment: a peak earlier than this after a pulse is not caused by it; onset gate: only for contractions without upstroke in the data
maxStimToPeak peak assignment	auto	peak assignment: auto = min(stimulus interval, 1 s), or a number of seconds
externalTrigger Stimuli	auto	stimulus times: MyoDish pulses of the channel, or the external trigger pulses of the status channel (external stimulator at the external controller unit: one chamber, any data channel). auto = external trigger pulses if the window has no MyoDish pulses

5 Tab Detection: contractions, locking and certainty

Contractions are peaks with a prominence above the threshold (auto threshold per channel and window, or a manual value in the main window). A channel counts as paced with at least pacedMinStimuli regular pulses in the window; then the typical amplitude is the median of the largest peaks (as many as pulses), the certainty uses the locking to the stimuli, and the rocker rules apply.

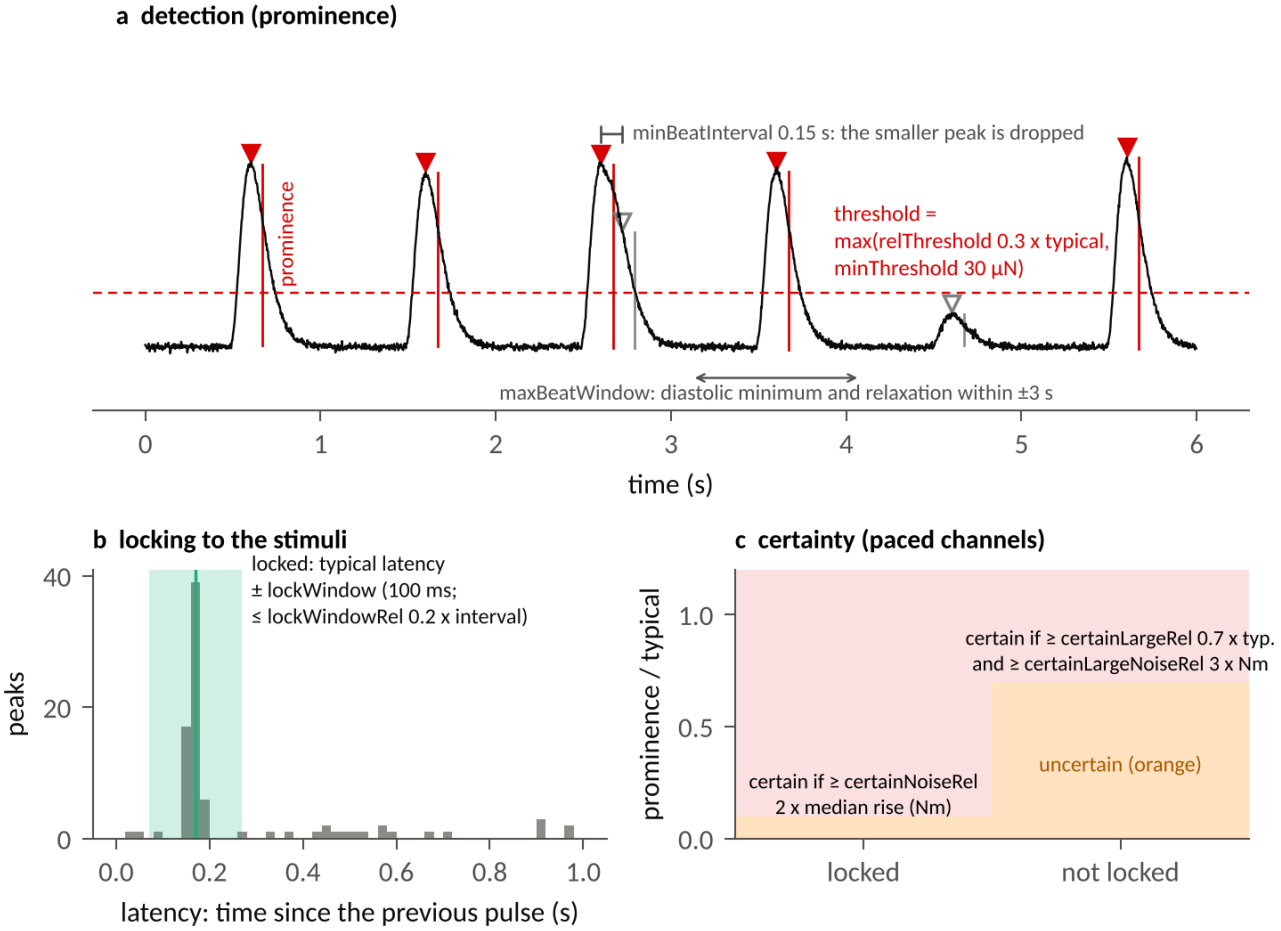


Figure 8. (a) Detection; (b) locking: the typical latency is the centre of the densest window of 2 x lockWindow; (c) certainty of paced contractions (uncertain = orange, counted only with high sensitivity).

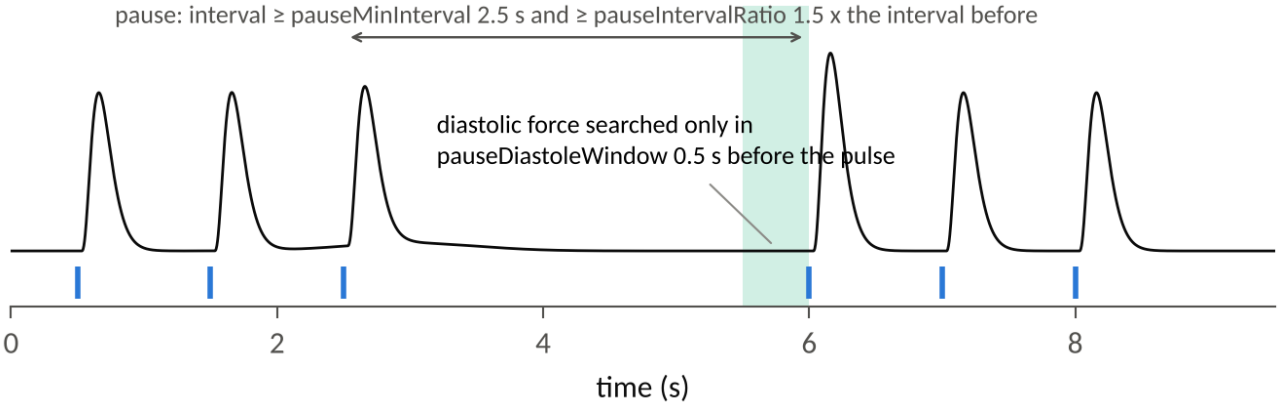
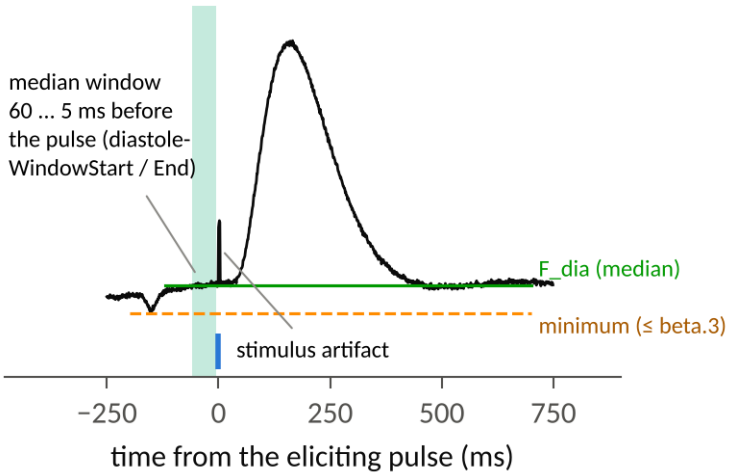


Figure 9. Stimulation pause: the diastolic minimum of the first contraction after the pause is searched only shortly before its pulse (diastolicLevel 'minimum').

Diastolic level F_dia (diastolicLevel, since 1.0.0-beta.4): the median of the unfiltered signal in a window diastoleWindowStart … diastoleWindowEnd (60 … 5 ms) before the eliciting pulse. The window ends before the stimulus artifact, and the median is not pulled down by noise, rocker movement or a short dip before the pulse, which lowered the former level (the last minimum of the filtered signal before the peak) and enlarged the

amplitude. Amplitude, diastolic force, the upstroke levels and the AUC refer to F_dia; the relaxation levels refer to the minimum after the peak. Contractions without a pulse (extra, unpaced) and contractions whose median is at or above the peak use the minimum before the peak. The former level is available as 'minimum'.

a diastolicLevel: median before the pulse



b level used

amplitude, diastolicForce	F_dia
upstroke (TTP, rise time)	F_dia
AUC	F_dia
relaxation (TTR, CD)	min. after the peak
no pulse (extra, unpaced)	min. before the peak
median ≥ peak	min. before the peak

Figure 10. (a) The median window before the pulse and the former minimum; (b) the level each parameter refers to.

If the onset follows the eliciting pulse by more than gateMax (longer gate of certain contractions), the median window ends before the onset instead of before the pulse, because the force at the pulse may still fall (incomplete relaxation). Besides stimToOnset, every contraction has stimToRise50, the delay from the pulse to 50 % of the amplitude on the upstroke; it depends less on the shape of the foot of the upstroke (slow or irregular onsets).

Inverted signal (invertedCheck, since 1.0.0-beta.5): in a paced channel, the median of the filtered signal is taken from invertedSkip (0.02 s) to min(1 s, median stimulus interval) after the pulses (at most 300 pulses, baseline 60 … 5 ms before each pulse). If this stimulus-triggered median falls at least invertedRatio (2) times as much as it rises, by at least invertedMinAmplitude (50 µN) and for at least invertedMinWidth (0.05 s) below half the fall, and if the signal itself deflects downwards (50th − 10th percentile ≥ invertedRatio × (90th − 50th percentile)), the force measurement is not valid (e.g. the magnet of the MyoDish out of position). The median and the percentiles are not misled by steps (chamber taken out), beats that are not locked to the pulses, or upright contractions just before the pulses. The contractions are then detected in the negated signal: rate and capture are analysed, all force and time parameters are NaN, and the column invertedSignal is true.

a longer gate of a certain contraction

b stimulus-triggered median (inverted)

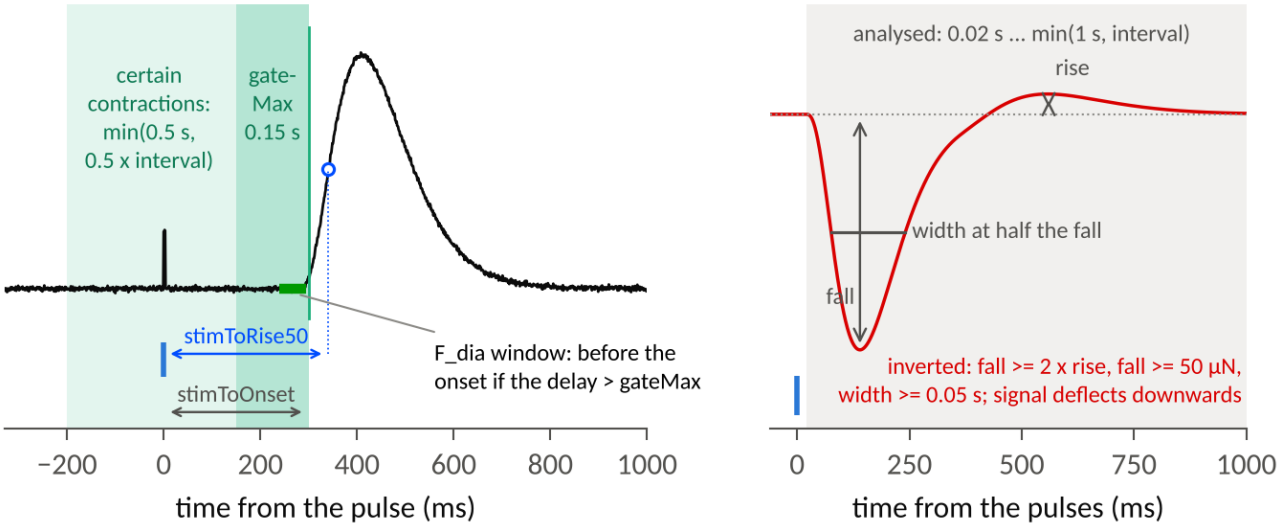


Figure 11. (a) Longer gate of a certain contraction 0.3 s after the pulse (0.5 Hz pacing), the median window of F_dia before the onset, stimToOnset and stimToRise50; (b) the stimulus-triggered median of an inverted channel and the criteria of invertedCheck.

Settings of the tab (* = the data are read again):

setting (GUI label)	default	meaning
relThreshold auto threshold	0.3 x typical	auto threshold: minimum prominence = this fraction of the typical contraction amplitude
minThreshold auto threshold	30 µN	auto threshold: never lower than this
minBeatInterval	0.15 s	minimum time between two contraction peaks (the more prominent one wins)
maxBeatWindow	3 s	at most this time before and after a peak belongs to one contraction (diastolic minimum, relaxation)
pacedMinStimuli	3 pulses	a channel counts as paced with at least this many regular pulses in the data window: typical amplitude = median of the largest peaks (as many as pulses), noise level before the stimuli, rocker rules, certainty by locking; otherwise: largest gap between the sorted prominences
pauseMinInterval	2.5 s	stimulation pause: stimulus interval at least this long …
pauseIntervalRatio	1.5 x interval before	… and at least this multiple of the interval before; then F_dia is searched only from pauseDiastoleWindow before the stimulus
pauseDiastoleWindow	0.5 s	after a stimulation pause, the diastolic minimum is searched only from this time before the stimulus (inf = off)
diastolicLevel Diastolic level (F_dia)	median before the pulse	level from which amplitude, upstroke levels (10 / 50 / 90 %), AUC and diastolic force are measured. median before the pulse: median of the unfiltered signal diastoleWindowStart … diastoleWindowEnd before the eliciting pulse (before the onset if it follows the pulse by more than gateMax; contractions without pulse, or a median above the peak: minimum); minimum: last minimum of the filtered signal before the peak (versions <= 1.0.0-beta.3). Relaxation levels: minimum after the peak (both)
diastoleWindowStart before the pulse	60 ms	diastolic level 'median before the pulse': window from this time before the eliciting pulse …
diastoleWindowEnd before the pulse	5 ms	… to this time before the pulse (stimulus artifact excluded)
lockWindow stimulus locking	100 ms	a peak is locked to the stimuli if its latency (time since the previous pulse) lies within +-lockWindow of the typical latency (centre of the densest window of 2 x lockWindow); certainty and rocker / artifact rules
lockWindowRel	0.2 x stimulus interval	at high rates the lock window is at most this fraction of the stimulus interval (certainty; rocker-only rule)
certainNoiseRel	2 x median rise	certainty, paced: a locked contraction is certain if its prominence is >= this multiple of the median rise of the signal before the stimuli
certainLargeRel	0.7 x typical	certainty, paced: a contraction that is not locked is certain if it is >= this fraction of the typical amplitude …
certainLargeNoiseRel	3 x median rise	… and >= this multiple of the median rise before the stimuli
certainUnpacedRel	0.5 x typical	certainty, unpaced channels: certain if >= this fraction of the typical amplitude
invertedCheck Inverted signal	on	paced channels: if the force falls after the stimuli (magnet out of position), the contractions are detected in the inverted signal (rate, capture) and the force and time parameters are NaN (column invertedSignal)
invertedRatio	2 x rise	inverted signal: the largest fall of the stimulus-triggered median is at least this multiple of its largest rise, and the signal deflects downwards (50th - 10th percentile >= this x (90th - 50th percentile))
invertedMinAmplitude	50 µN	inverted signal: the fall of the stimulus-triggered median is at least this large
invertedMinWidth	50 ms	inverted signal: the stimulus-triggered median lies below half its fall for at least this time (no stimulus artifact)
invertedSkip	20 ms	inverted signal: the stimulus-triggered median starts this time after the pulse (stimulus artifact)

6 Tab Noise / artifacts: noise level and artifact gap rule

The noise level N of a paced channel is measured where no contraction is expected: in a window before each pulse. It is used by the certainty and by the rocker rules. The artifact gap rule raises the auto threshold above a separate cluster of small peaks that are not locked to the stimuli and occur while the rocker moves. "More than by chance": the fraction of locked peaks exceeds the chance level (2 x lockWindow / stimulus interval) + chanceMargin, at most chanceMax.

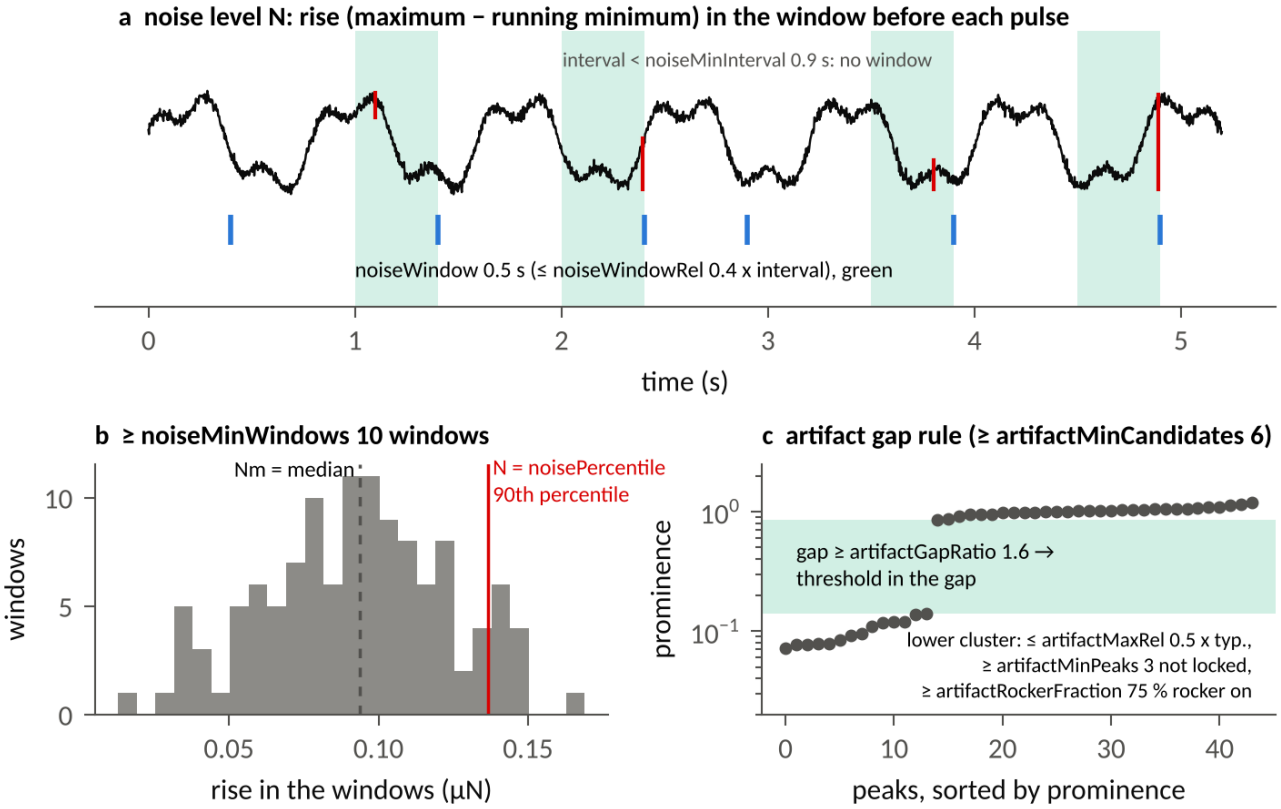


Figure 12. (a, b) Noise level N and its median Nm; (c) artifact gap rule.

Settings of the tab (* = the data are read again):

setting (GUI label)	default	meaning
rockerArtifacts Rocker artifact rules	on	paced channels, auto threshold: peaks of the rocker movement that are not locked to the stimuli are not counted (artifact gap and rocker rules)
noiseMinInterval	0.9 s	noise level N: only windows before stimuli that follow an interval of at least this length
noiseWindow	0.5 s	noise level N: window before each such stimulus (at most noiseWindowRel x interval); rise = maximum minus the running minimum of the signal
noiseWindowRel	0.4 x interval	noise window at most this fraction of the stimulus interval
noisePercentile	90 %	noise level N = this percentile of the rises (the median of the rises is used for the certainty)
noiseMinWindows	10 windows	noise level only with at least this many windows (otherwise unknown)
artifactMinCandidates	6 peaks	artifact gap rule (raises the threshold): needs at least this many peaks above the auto threshold
artifactGapRatio	1.6 x	artifact gap rule: the largest ratio between two consecutive sorted prominences below the typical amplitude must be >= this value (a clear gap between artifacts and contractions)
artifactMaxRel	0.5 x typical	artifact gap rule: all peaks of the lower cluster are <= this fraction of the typical amplitude
artifactMinPeaks	3 peaks	artifact gap rule: at least this many peaks of the lower cluster are not locked to the stimuli
artifactRockerFraction	75 %	artifact gap rule: at least this fraction of the lower cluster occurs while the rocker moves
chanceMargin	20 %	locked 'more than by chance': fraction locked > chance level (2 x lockWindow / stimulus interval) + this margin …
chanceMax	60 %	… the limit is at most this fraction (artifact gap rule, partial capture of the rocker rules)

7 Tab Rocker rules: peaks of the rocker movement

Paced channels with the rocker moving during at least rockerRuleMinFraction of the window. Rule b: a channel without contractions (few peaks locked, small typical amplitude or peaks at the rocker rhythm): rocker / noise peaks are removed, locked peaks that are clearly above the noise stay. Rule c: small peaks that are not locked are removed. Without a noise level (high rates), the rocker-only rule uses the peak intervals (1 or ½ rocker period).

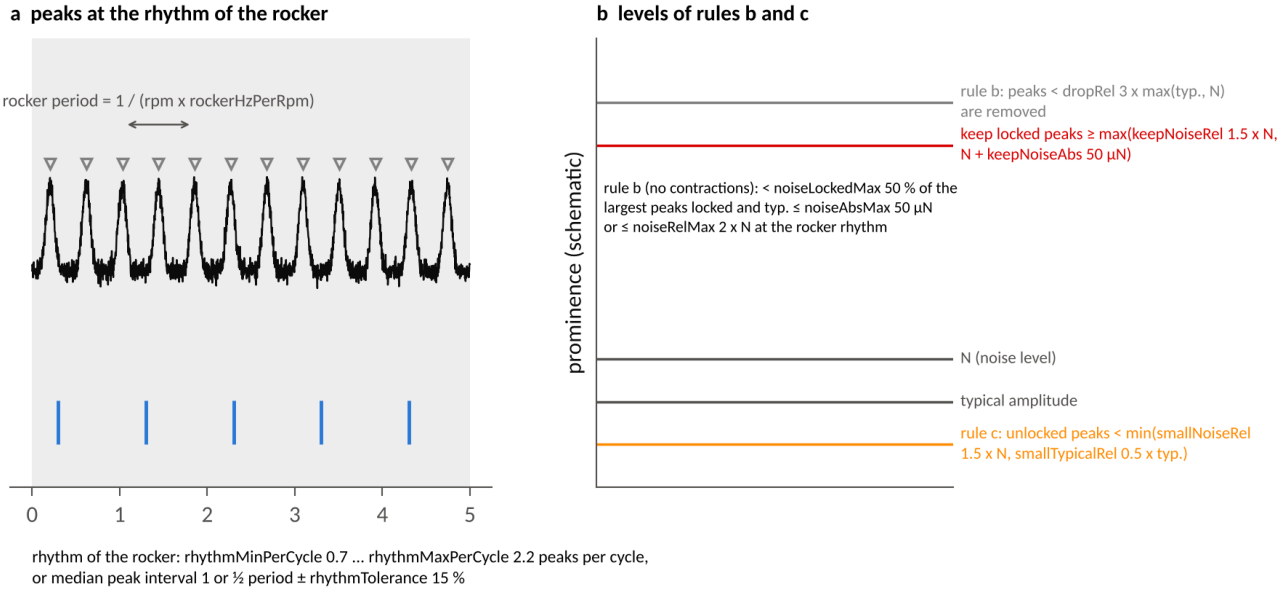


Figure 13. (a) Peaks at the rhythm of the rocker; (b) the levels of rules b and c (schematic order).

Settings of the tab (* = the data are read again):

setting (GUI label)	default	meaning
rockerRuleMinFraction	50 % of the time	rocker rules (b, c): only if the rocker moves during at least this fraction of the data window
noiseLockedMax	50 %	rule b (no contractions): fewer than this fraction of the largest peaks (as many as pulses) locked …
noiseAbsMax	50 µN	… and the typical amplitude <= this value …
noiseRelMax	2 x N	… or <= this multiple of the noise level N with the peaks at the rhythm of the rocker: rocker / noise peaks
rhythmMinPerCycle	0.7 peaks / cycle	rhythm of the rocker: at least this many peaks per rocker cycle (rule b) …
rhythmMaxPerCycle	2.2 peaks / cycle	… and at most this many, or a median peak interval of 1 or 1/2 rocker period …
rhythmTolerance	15 % of the period	… within this tolerance (1/2 period: half of it)
keepNoiseRel	1.5 x N	rule b: locked peaks >= max(keepNoiseRel x N, N + keepNoiseAbs) stay (slices that answer only some stimuli) …
keepNoiseAbs	50 µN	… (absolute distance from the noise level) …
keepMinPeaks	3 peaks	… if there are at least this many of them …
keepMinFraction	5 % of the pulses	… and at least this fraction of the pulses, and more of them are locked than by chance
dropRel	3 x max(typical, N)	rule b: peaks below this multiple of max(typical amplitude, N) are removed (larger peaks are no rocker artifacts)
smallNoiseRel	1.5 x N	rule c: peaks not locked to the stimuli below min(smallNoiseRel x N, smallTypicalRel x typical) are removed …
smallTypicalRel	0.5 x typical	… (peaks at the rhythm of the rocker: smallRhythmTypicalRel instead) …
smallRhythmMinPerCycle	0.5 peaks / cycle	… rhythm of the rocker for rule c: at least this many peaks per rocker cycle …
smallRhythmTypicalRel	1 x typical	… then the limit is min(smallNoiseRel x N, this fraction of the typical amplitude)
rockerOnlyMinIntervals	10 intervals	N unknown (high rates): peaks at the rhythm of the rocker need at least this many peak intervals …
rockerOnlyFraction	60 %	… of which at least this fraction are 1 or 1/2 rocker period …
rockerOnlyTolerance	10 % of the period	… within this tolerance (1/2 period: half of it) …
captureTolerance	5 %	… and the median interval is no 1:1 or 2:1 multiple of the stimulus interval within this tolerance (capture at the rocker period cannot be told apart)

8 Tab Signal: averaging, filters and spike removal

The raw samples are averaged (downsampling, 2: 400 → 200 Hz), filtered with a moving median and a moving mean. Before the averaging, spike artifacts (e.g. a chamber taken out or put in) are replaced by a line.

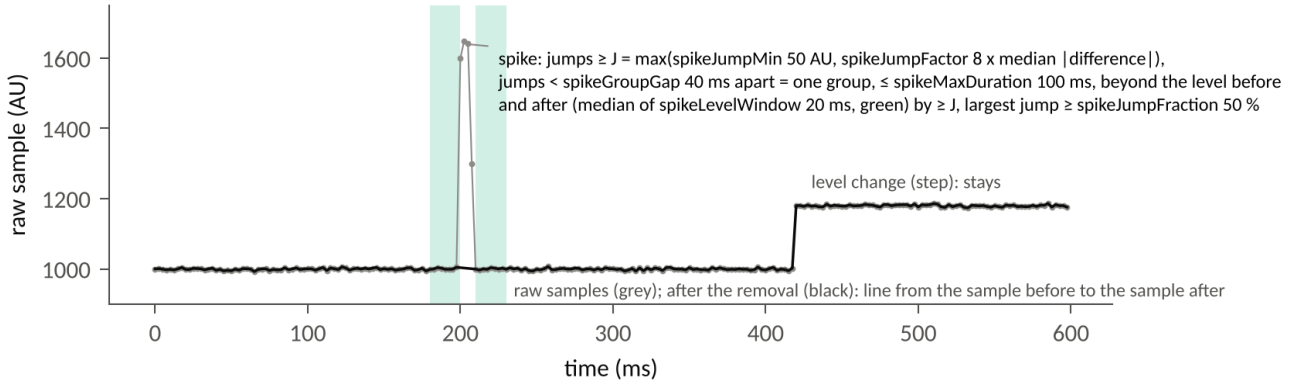


Figure 14. Spike removal. A level change (step) is no spike and stays.

Settings of the tab (* = the data are read again):

setting (GUI label)	default	meaning
downsampling * raw samples per value	2 samples	n raw samples are combined (2: mean, 400 Hz -> 200 Hz; n > 2: median); the window is read again
medianFilterMs Moving median	50 ms	moving median of the force signal (0 = off)
meanFilterMs Moving mean	25 ms	moving mean after the median (0 = off)
spikeRemoval * Remove spike artifacts	on	spike artifacts (e.g. chambers taken out / put in) are replaced by a line before the averaging of the raw samples (the window is read again)
spikeJumpMin *	50 AU	spike: a jump between two raw samples >= J = max(spikeJumpMin, spikeJumpFactor x median of the non-zero differences of the channel) ...
spikeJumpFactor *	8 x noise	... (noise level of the channel) ...
spikeGroupGap *	40 ms	jumps closer than this form one group ...
spikeMaxDuration *	100 ms	... that lasts at most this long ...
spikeLevelWindow *	20 ms	... and goes beyond the level before and after it (median of this time) by >= J ...
spikeJumpFraction *	50 %	... with the largest jump >= this fraction of its largest deviation (abrupt; a contraction rises over many samples)
spikeCoincidence *	10 ms	within this time of a spike of another channel ...
spikeCoincidenceFactor *	0.5 x J	... the jump threshold is max(spikeJumpMin, this factor x J) (spikes in several channels)

9 Tab Rocker filter: rocker state and periodic artifact

The rocker state comes from bit 15 of the status channel or from the rockerSpeed entries of the log file. The rocker filter (main window: remove rocker artifact) fits a periodic artifact with the rocker frequency to the samples between the contractions and subtracts it; the frequency is refined within a band around rpm x rockerHzPerRpm. Blocks are accepted only if they cover the rocker cycle and their fit is consistent (R^2, size); rejected blocks borrow the artifact of a neighbouring accepted block.

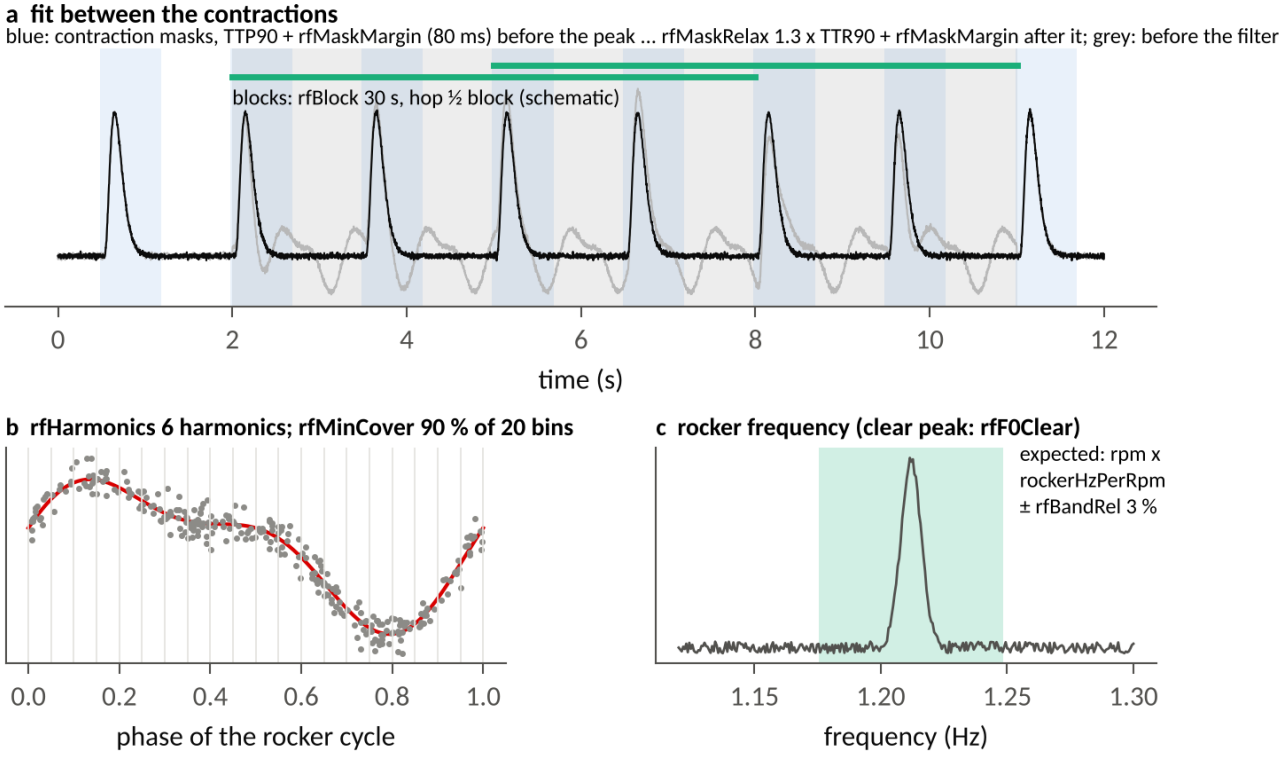


Figure 15. Rocker filter: (a) masks and blocks; (b) the periodic artifact over the phase of the rocker cycle; (c) search of the rocker frequency.

Settings of the tab (* = the data are read again):

setting (GUI label)	default	meaning
rockerSource * Rocker state from	auto	auto: bit 15 of the status channel; the 'rockerSpeed' entries of the log file if that bit is missing or there is no status channel; the file is read again
rockerLogDelay *	0.27 s	delay of the rocker movement after a 'rockerSpeed' entry of the log file (rocker state from the log file)
rockerHzPerRpm	0.0202 Hz / rpm	rocker frequency per rpm setting (two setups: 0.02020 and 0.02025 Hz/rpm); rocker rules and rocker filter
rfBandRel frequency search	3 %	rocker filter: the rocker frequency is searched within +-this fraction of the expected frequency
rfBandNoLog	0.4 – 2.2 Hz (from to)	rocker filter: search band without rocker speed in the log file
rfHarmonics	6 harmonics	rocker filter: number of harmonics of the periodic artifact
rfBlock	30 s	rocker filter: block length of the fit (hop = half a block)
rfMinRun	3 s	rocker filter: shorter rocker-on periods are not corrected
rfKnot	2 s	rocker filter: knot distance of the baseline (slow drift) fitted together with the artifact
rfSmoothBaseline	0.1	rocker filter: smoothness penalty of the baseline
rfRidge	0.001	rocker filter: ridge penalty of the harmonics
rfMinCover	90 %	rocker filter: a block is fitted only if this fraction of the 20 phase bins of the rocker cycle has >= 5 samples between the contractions
rfMinR2	0.2	rocker filter: minimum R2 of a block fit
rfMaxSizeRel	1.6 x reference	rocker filter: artifact size of a block at most this multiple of the reference size
rfMinRefBlock	10 s	rocker filter: reference blocks are at least this long
rfMaxBorrow	60 s	rocker filter: a block without own fit uses the artifact of a neighbouring block of the same rocker period up to this distance
rfF0Block	120 s	rocker filter, frequency estimate: blocks of at most this length …
rfF0Blocks	6 blocks	… at most this many per channel and rocker speed …
rfF0Clear	10 %	… and the peak of the fit must be clear within +-this fraction
rfExtendBlock	90 s	rocker filter: blocks without full coverage of the rocker cycle are retried with this length
rfMaskRelax contraction masks	1.3 x TTR90	rocker filter: the artifact is fitted to the samples between contractions; a contraction is masked from TTP90 + rfMaskMargin before its peak to rfMaskRelax x TTR90 + rfMaskMargin after it (medians of the channel)
rfMaskMargin	80 ms	rocker filter: margin of the contraction masks before and after a contraction
rfMaskMax paced masks	1.2 s	rocker filter, paced channels (first pass): masked from 20 ms before each pulse to min(next pulse - 50 ms, this time) after it

10 Tab Protocols: grouping, FFR steady state and PRP reference

Protocols (FFR, post-rest potentiation) group the contractions by the stimulation. Pacing frequencies are rounded to frequencyResolution (0.1 Hz) at ≥ 1 Hz and to frequencyResolutionLow (0.05 Hz) below 1 Hz: 0.99, 1.0 and 1.001 Hz are one group "1 Hz", 0.74 and 0.75 Hz one group "0.75 Hz". Pauses that differ by at most pauseTolerance are one pause length; PRP15 / 30 / 60 are the means of the pauses within this tolerance of 15 / 30 / 60 s.

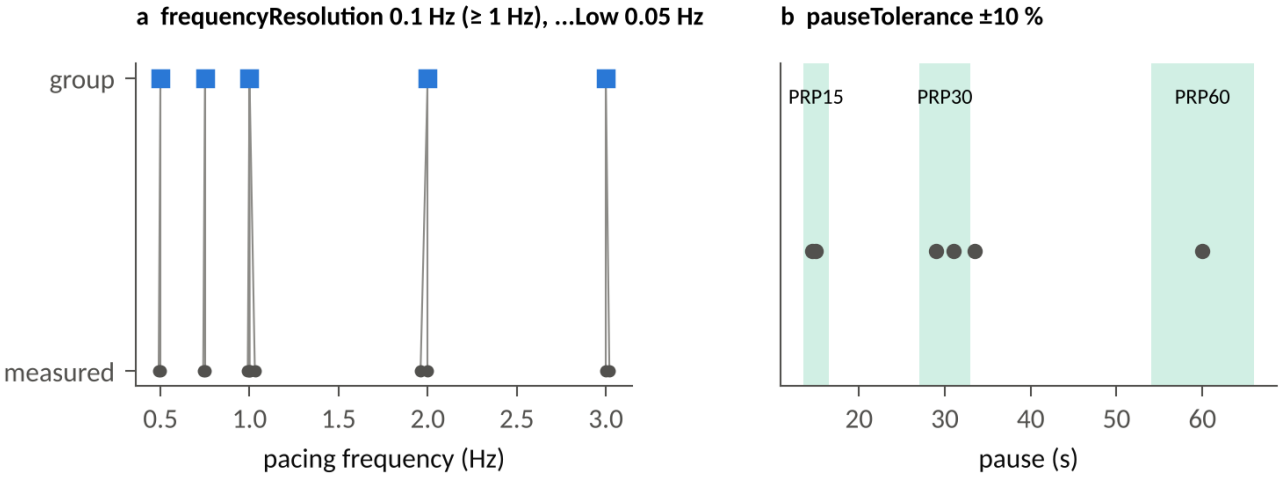


Figure 16. (a) Groups of the pacing frequency; (b) pauses that count for PRP15 / 30 / 60.

FFR steps (since 1.0.0-beta.4): each pacing frequency is summarized from its steady state, the last steadyStateBeats (10) contractions of the longest run of consecutive stimuli at this frequency. Other runs at the same frequency (e.g. the basic pacing before and after the protocol) are not used. Within the run, only contractions whose previous and next stimuli elicited a contraction, too, count (longest such sequence): partial capture (e.g. 2:1) gives no steady state (note). With the rocker 'stopped' (main window), only contractions with the rocker at rest are used; a step without any (rocker not stopped during this step) is summarized from its contractions with the rocker moving (ffrRockerFallback) and noted. Every group gets the coefficient of variation of the amplitudes (amplitude_CV) and the flag irregular (amplitude_CV > irregularCV, e.g. conduction block, alternans or extra beats); steps with fewer than minGroupBeats included contractions are listed in the notes.

Post-rest potentiation (prpReference): by default, each post-rest contraction is divided by the median of the last prpReferenceBeats (6) contractions of the pacing before its pause. This pacing loads the sarcoplasmic reticulum before the pause, so slow changes of the amplitude during the protocol (run-down, recovery after an earlier pause) do not enter the ratio. 'firstTrain': mean of the contractions of the pacing before the first pause (one reference for all pauses); 'steady': mean of the steady group (versions ≤ 1.0.0-beta.3).

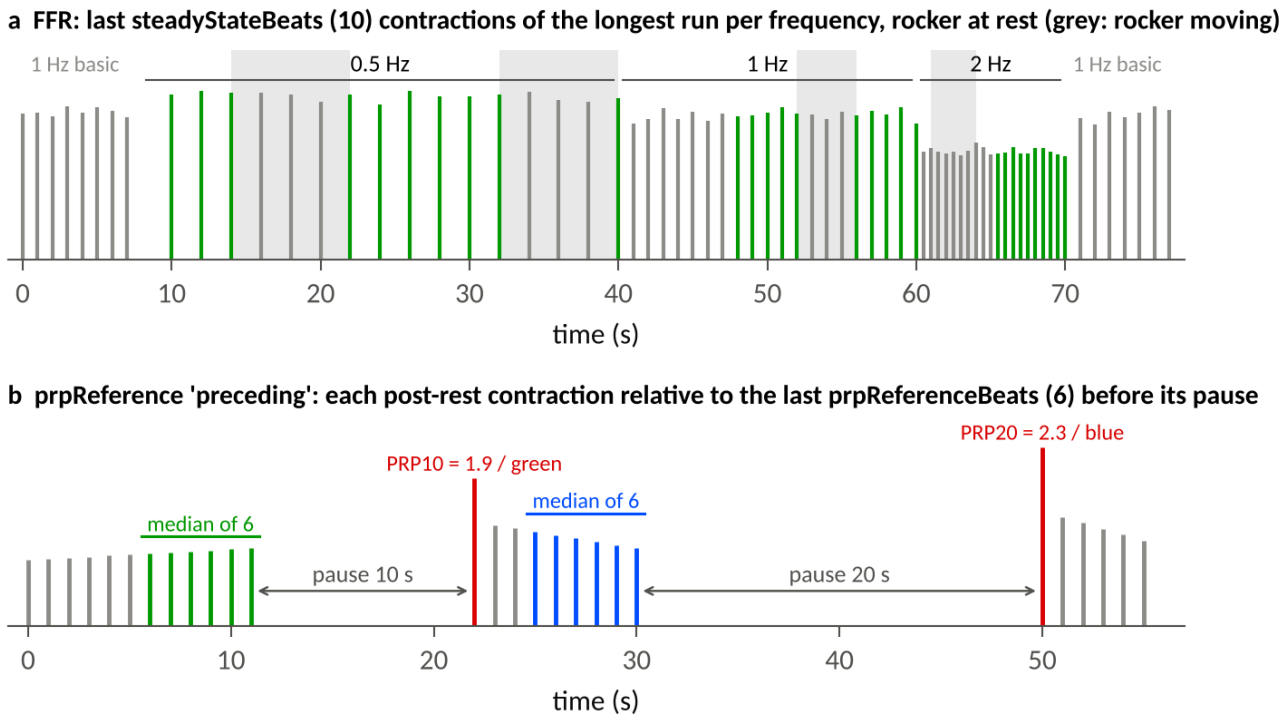


Figure 17. (a) Contractions that summarize the FFR steps (green); (b) the reference of each post-rest contraction.

Settings of the tab (* = the data are read again):

setting (GUI label)	default	meaning
frequencyResolution >= 1 Hz	0.1 Hz	grouping by pacing frequency (FFR): stimulus intervals within 2 % form a group, its frequency (1 / median interval) is rounded to this step at >= 1 Hz; groups with the same rounded frequency are one group (0.99, 1.0 and 1.001 Hz = 1 Hz). FFR results: the groups at 0.5, 1, 2 and 3 Hz
frequencyResolutionLow < 1 Hz	0.05 Hz	... below 1 Hz the frequency is rounded to this step (0.49 and 0.5 Hz = 0.5 Hz, 0.74 and 0.75 Hz = 0.75 Hz)
pauseTolerance pause length	10 %	grouping by pause length (post-rest potentiation): pauses (stimulus interval - steady interval) within this fraction of the shortest pause of a set are one pause length (same group value and label, numbered #n); PRP15 / 30 / 60: mean of the pauses within this fraction of 15 / 30 / 60 s
steadyStateBeats FFR steps	10 contractions	FFR protocols: each pacing frequency is summarized from its steady state: the last this many contractions of the longest run of consecutive stimuli at this frequency (other runs, e.g. the basic pacing before and after the protocol, are not used), of its longest sequence of captured stimuli whose previous and next stimuli are captured, too (partial capture, e.g. 2:1, gives no steady state). 0 = all contractions of the frequency (versions <= 1.0.0-beta.3)
ffrRockerFallback FFR: steps without rocker stop	on	FFR protocols with rocker 'stopped' (default of protocols): contractions with the rocker at rest are used (more accurate); a step without such contractions is summarized from the contractions with the rocker moving (note)
prpReference PRP reference	last beats before each pause	post-rest potentiation: reference amplitude of each post-rest contraction. last beats before each pause: median of the last prpReferenceBeats contractions of the pacing before this pause (this pacing loads the SR); train before the first pause: mean of the contractions of the pacing before the first pause of the protocol; all steady contractions: mean of the steady group (versions <= 1.0.0-beta.3). Reference contractions independent of the rocker state
prpReferenceBeats	6 contractions	PRP reference 'last beats before each pause': number of contractions before the pause (median)
irregularCV irregular groups	15 %	a group (e.g. FFR step) is flagged irregular (summary column irregular) if the coefficient of variation of the amplitudes of its included contractions (population SD / mean, column amplitude_CV) exceeds this value: conduction block, alternans or extra beats
minGroupBeats few contractions	5 contractions	FFR protocols: a note lists the frequencies summarized from fewer included contractions (0 = no note)

11 Tab Export

The pulse table (sheet / file *pulses*) lists every stimulus pulse of the analysed channels and ranges; prePulses / postPulses of the contraction table list at most pulseTextMax pulses.

Settings of the tab (* = the data are read again):

setting (GUI label)	default	meaning
pulseTable Pulse table	on	exports and All channels -> file: table of all stimulus pulses (sheet / file pulses: pulse ID, time, extra pulse, current, durations, outcome, role, contraction, phase)
pulseTextMax	10 pulses	prePulses / postPulses: at most this many pulses (those nearest to the eliciting pulse), then '&+<number of the others>'

12 Fixed in this version

Purely numerical details (e.g. 20 phase bins of the rocker cycle, the 50-Hz decimation of the frequency search, the 2 % clustering of stimulus intervals before the rounding) and the criteria of the protocol results (captured group: at most max(1, 10 %) stimuli without contraction; stimulation thresholds at 10, 50, 95 and 99 % of the largest amplitude; the S2 response analysis of the refractory period) are not settings. They are described in the help of mda_groupBeats and mda_protocolResults.